

BFO 2020 Participation Axioms

Participates in and has participant are inverse relations [xjr-1]

$$\forall t,a,b (\text{participatesIn}(a,b,t) \leftrightarrow \text{hasParticipant}(b,a,t))$$

At every time a process exists it has a participant [trl-1]

$$\forall p,t (\text{instanceOf}(p,\text{process},t) \rightarrow \exists c \text{ participatesIn}(c,p,t))$$

Participates in is disjunctive on third argument, a temporal region [yjm-1]

$$\forall p,q,r,s (\text{participatesIn}(p,q,r) \wedge \text{temporalPartOf}(s,r) \rightarrow \text{participatesIn}(p,q,s))$$

If c participates in p at t and p occupies temporal region r then t is part of r [kxe-1]

$$\forall c,p,r,t (\text{occupiesTemporalRegion}(p,r) \wedge \text{participatesIn}(c,p,t) \rightarrow \text{temporalPartOf}(t,r))$$

Participates in is time indexed and has domain: independent continuant or specifically dependent continuant or generically dependent continuant but not spatial region and range: process [ild-2]

$$\begin{aligned} \forall a,b,t (\text{participatesIn}(a,b,t) \\ \rightarrow (\text{instanceOf}(a,\text{independentContinuant},t) \\ \vee \text{instanceOf}(a,\text{specificallyDependentContinuant},t) \\ \vee \text{instanceOf}(a,\text{genericallyDependentContinuant},t)) \\ \wedge \neg \text{instanceOf}(a,\text{spatialRegion},t) \wedge \text{instanceOf}(b,\text{process},t) \\ \wedge \text{instanceOf}(t,\text{temporalRegion},t)) \end{aligned}$$

At every time a specific dependent s participates in a process p there's a part of that time, during which there's an independent continuant that s depends on, and that participates in p at that time [cgn-1]

$$\begin{aligned} \forall \text{sdc},p,t (\text{instanceOf}(\text{sdc},\text{specificallyDependentContinuant},t) \wedge \text{participatesIn}(\text{sdc},p,t) \\ \rightarrow \exists tp,ic (\text{instanceOf}(tp,\text{temporalRegion},tp) \wedge \text{temporalPartOf}(tp,t) \\ \wedge \text{instanceOf}(ic,\text{independentContinuant},tp) \\ \wedge \neg \text{instanceOf}(ic,\text{spatialRegion},tp) \wedge \text{specificallyDependsOn}(\text{sdc},ic) \\ \wedge \text{participatesIn}(ic,p,tp))) \end{aligned}$$

If a generically dependent continuant participates in a process p then, if it is concretized as a process, that process is part of p, and if concretized as an sdc then the bearer of that sdc participates in the process [fmm-1]

$$\begin{aligned} \forall \text{gdc},p,t (\text{instanceOf}(\text{gdc},\text{genericallyDependentContinuant},t) \wedge \text{participatesIn}(\text{gdc},p,t) \\ \rightarrow \exists tp,b (\text{temporalPartOf}(tp,t) \wedge \text{concretizes}(b,\text{gdc},tp) \\ \wedge ((\text{instanceOf}(b,\text{specificallyDependentContinuant},tp) \\ \wedge (\exists ic (\text{specificallyDependsOn}(b,ic) \wedge \text{participatesIn}(ic,p,tp)))) \\ \vee (\text{occurrentPartOf}(b,p) \wedge \text{existsAt}(b,tp)))))) \end{aligned}$$

